Chapter 7 Sec 88

Jordan's Lemma

Jordan's Lemma serves as an aid in evaluating integrals of the type:

$$\int_{-\infty}^{\infty} f(x)e^{iax} = \int_{-\infty}^{\infty} f(x)\cos(ax) + i\int_{-\infty}^{\infty} f(z)\sin(z)$$

Jordan's Lemma

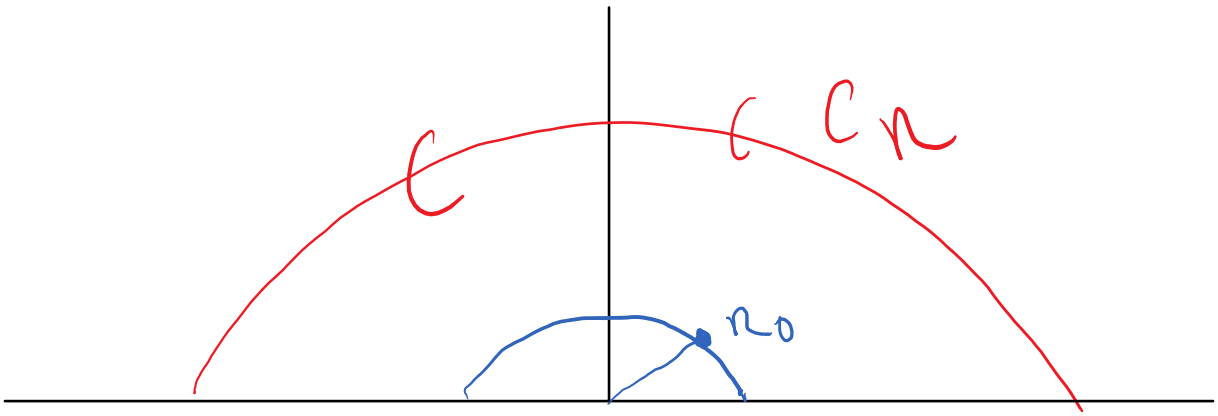
Suppose that

a) A function $f(z)$ is analytic at all points in the upper half plane $y \geq 0$ that are exterior to the a circle $|z| = R_0$ where R_0 can take on the value of zero.
b) C_R denotes a semicircle $z = Re^{i\theta}(0 \leq \theta \leq \pi)$ whre $R > R_0$
c) For all points z on C_R, there is a positive constant M_R such that:

$$|f(z)| \leq M_R \quad and \quad \lim_{r\to\infty} M_R = 0$$

Then for every positive constant a,

$$\lim_{R\to\infty} \int_{C_R} f(z)e^{iaz}dz = 0$$



What this theorem is giving us is a criteria on $f(z)$, $|f(z)| \leq M_R$ that tells us when the integral $\int_{C_r} f(z)e^{iaz}dz$ is zero rather that having to estimate the integral it's self. This means that for our formula:

$$\int_{-\infty}^{\infty} f(x)e^{iaz}dx = 2\pi i \sum_{k=1}^{n} Res_{z=z_k} f(z)e^{iaz} - \lim_{R\to\infty} \int_{C_R} f(z)e^{iaz}dz$$

We just need to bound $f(z)$ instead of the the whole rightmost integral to show that that it is zero.

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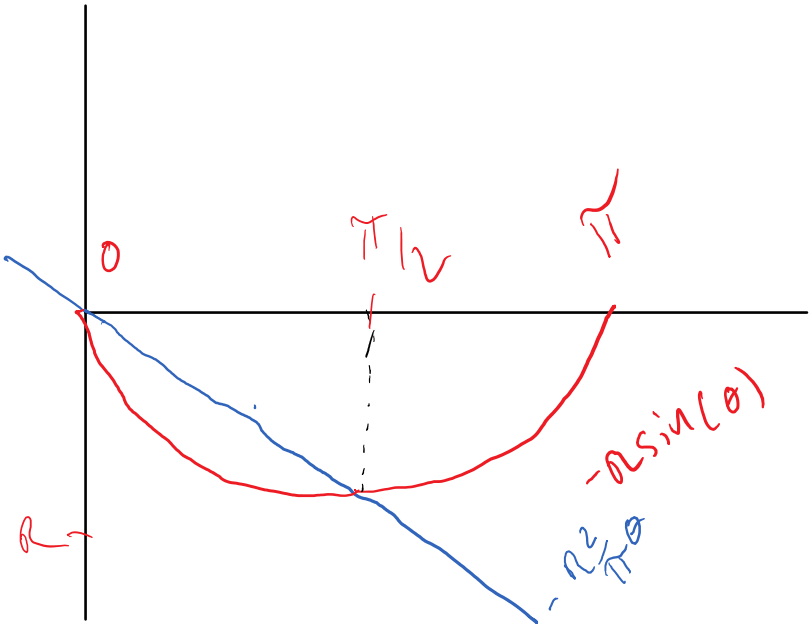
$$|f(z)| \leq M_R \quad and \quad \lim_{r\to\infty} M_R = 0$$

Then for every positive constant a,

$$\lim_{R\to\infty} \int_{C_R} f(z)e^{iaz}dz = 0$$

Proof: To prove this theorem we will first need to show Jordan's inequality $\int_0^{\pi} e^{-Rsin(\theta)}d\theta < \frac{\pi}{R}$

Consider the graphs of $y = -\mathrm{Rsin}(\theta), y = -R\frac{2}{\pi}\theta$



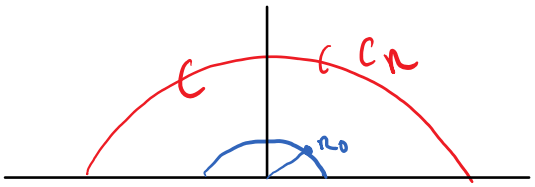
Clearly from this we can say

$$\int_0^{\frac{\pi}{2}} e^{-R\sin(\theta)}d\theta \leq \int_0^{\frac{\pi}{2}} e^{-\frac{2R\theta}{\pi}}d\theta$$

Now we just need to integrate and estimate $\int_0^{\frac{\pi}{2}} e^{-\frac{2R\theta}{\pi}}d\theta$ and use symmetry properties of sin around $x = \frac{\pi}{2}$

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So we now have that $\int_0^{\pi} e^{-Rsin(\theta)} d\theta < \frac{\pi}{R}$



So now let's just estimate $\int_{C_R} f(z)e^{iaz} dz$ Integrating along the contour $z(\theta) = Re^{i\theta}$ $0 \leq \theta \leq \pi$

Keeping in mind that $|f(z)| \leq M_R$ and $\lim_{(R\to\infty)} M_R = 0$ and that $\left|e^{iaRe^{i\theta}}\right| \leq e^{-aRsin(\theta)}$

6. Use Jordan's lemma to compute the principal value

$$PV \int_{-\infty}^{\infty} \frac{xe^{i\omega x}}{x^2+1} dx$$

where $\omega > 0$

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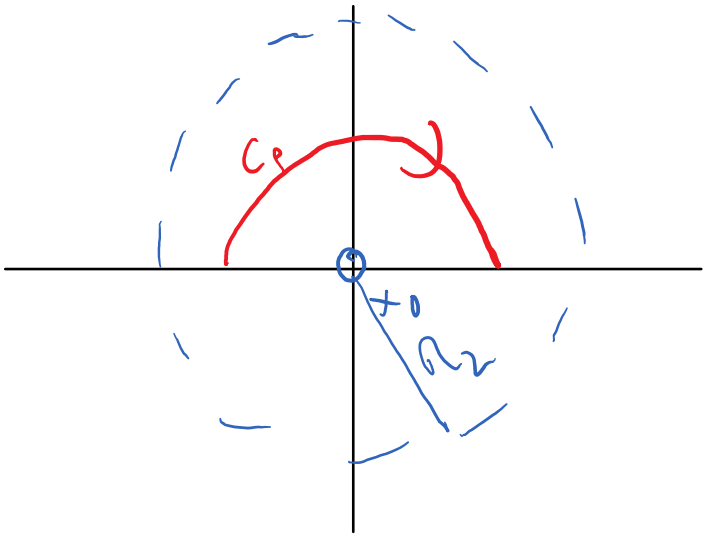
An Intended Path

Theorem Suppose that

a) A function $f(z)$ has a simple pole at a point $z = x_0$ on the real axis, with a Laurent series representation in a punctured disk $0 < |z - x_0| < R_2$ and with residue B_0
b) C_ρ denotes the upper half of a circle $|z - x_0| = \rho$, where $\rho < R_2$ and it is negatively oriented

Then

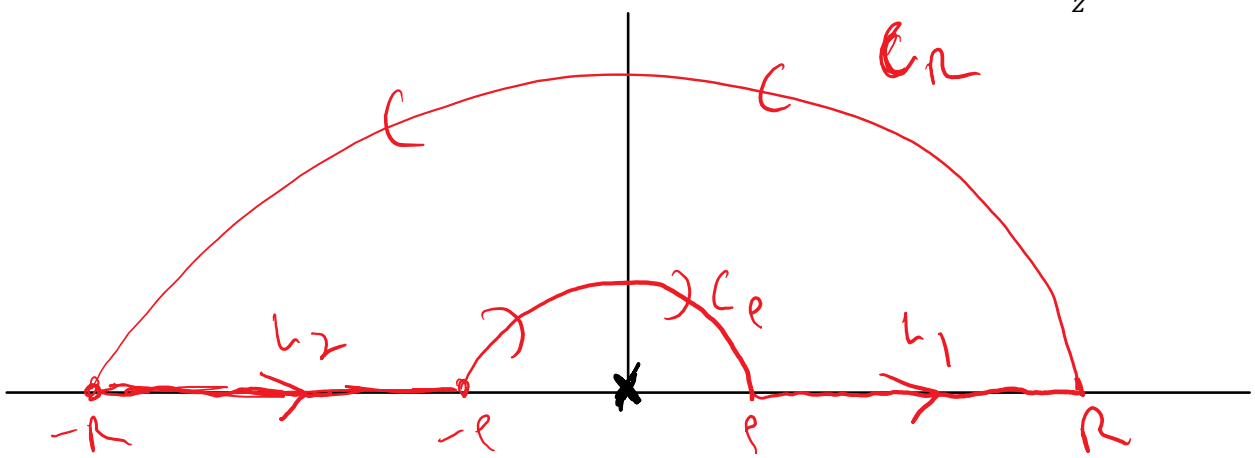
$$\lim_{\rho \to 0} \int_{C_\rho} f(z)dz = -B_0 \pi i$$



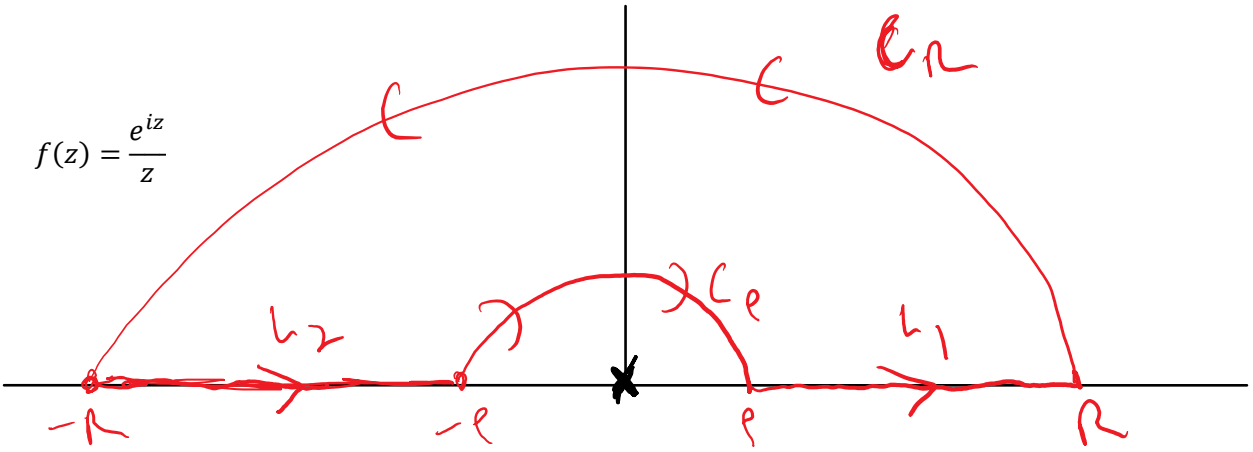
What this theorem allows us to do is to still be able to compute real improper integrals even when there is a singular point on the real axis!

Example: evaluate $\int_0^\infty \frac{\sin(x)}{x} dx$ This is called Dirichlet's integral .

We will find it by integrating $f(z) = \frac{e^{iz}}{z}$ *on the contour below*



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The Cauchy Goursat theorem says that that the integral around the sum of these contours is zero

Solving for the two integrals on L_1 and L_2 going along $-L_2$ and realizing they are just on the real axis gives .

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So we now have that $$2i\int_{\rho}^{R}\frac{\sin(r)}{r}dr=-\int_{c\rho}\frac{e^{iz}}{z}dz-\int_{C_R}\frac{e^{iz}}{z}dz$$

If we write out the Laurent Series for $\frac{e^{iz}}{z}$ using it's taylor series we can find the residue and see that it is a simple pole. (pole of order 1) satisfying our intended path theorem on C_ρ

Further we can examine $\left|\frac{1}{z}\right|\leq\frac{1}{R}=M_r$ and satisfies Jordan's Lemma

As a result

$$\lim_{\rho\to 0}\lim_{R\to 0}\ 2i\int_{\rho}^{R}\frac{\sin(r)}{r}dr=\lim_{\rho\to 0}\lim_{R\to 0}\ -\int_{c\rho}\frac{e^{iz}}{z}dz-\lim_{\rho\to 0}\lim_{R\to 0}\int_{C_R}\frac{e^{iz}}{z}dz$$